

1. **How did React manage the form data?**

React managed the form data using the useState hook. All input fields were stored in a single formData state object, and the handleChange function updated the appropriate field whenever the user entered information. When the form was submitted, React validated the inputs and sent the data to the PHP backend using fetch() and URLSearchParams.

2. **Why are controlled components useful?**

Controlled components ensure that React always controls the input values through state. This made it easy to perform client-side validation, display error messages, clear the form after a successful submission, and keep the user interface synchronized with the application state.

3. **How did PHP process the submitted information?**

PHP received the form data through a POST request, trimmed the input values, validated that all required fields were completed and that the email contained an "@", then inserted the record into the database using a prepared SQL statement. Finally, it returned a JSON response indicating success or failure.

4. **How did PHP connect to MySQL?**

PHP connected to the MySQL database using the MySQLi extension with the database host, username, password, and database name. After establishing the connection, it executed prepared SQL statements to insert and retrieve student records.

5. **What challenges did you encounter?**

The biggest challenges were configuring communication between React and PHP, handling CORS, ensuring both client-side and server-side validation worked correctly, and debugging database connection or API errors. Another challenge was making sure the student list refreshed automatically after a successful registration without reloading the page.